

Author Contribution Statement

Physics-Constrained Adaptive Slope for Differential Protection: A Projected-Gradient Law with Lyapunov Stability Guarantees

CRedit Taxonomy

Contribution	A. V. Eluvathingal	R. K. Prasad
Conceptualisation	Lead	Supporting
Formal analysis	Lead	Supporting
Funding acquisition	—	Lead
Investigation	Lead	—
Methodology	Lead	Supporting
Project administration	—	Lead
Software	Lead	—
Supervision	—	Lead
Validation	Lead	Supporting
Visualisation	Lead	—
Writing — original draft	Lead	—
Writing — review & editing	Supporting	Lead

Plain-Language Statement (for portal upload)

Arun V. Eluvathingal: Identified and formalised the slope adaptation problem as a constrained online parameter estimation task; derived the physics-feasible set $\mathcal{K}(\kappa_n)$ from the LM condition number; designed the projected-gradient adaptive law (Algorithm 1); carried out the complete Lyapunov stability analysis including Theorem 1, Lemma 1, and Corollary 1; implemented all simulations on the 33 kV mixed-IBR test feeder (2400 scenarios); produced all figures and tables; wrote the manuscript draft.

R. K. Prasad: Supervised the research programme; provided guidance on the connection between the Lyapunov framework and the SAMBPS architecture; reviewed and edited the final manuscript; secured project funding.

Conflict of interest: The authors declare no conflict of interest.

Code availability: MATLAB/Simulink simulation files and the projected-gradient implementation will be deposited in the IIT Madras institutional repository upon acceptance.